

RIVERSDALE, South Africa

WMO: 689260

Lat: 34.0958S

Lon: 21.2517E

Elev: 499

StdP: 14.43

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.2	40.7	31.7	26.6	58.9	34.2	29.4	54.5	13.8	60.6	12.1	60.6	0.9	250	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	90.5	68.6	87.0	67.8	84.1	66.8	71.4	84.4	70.1	81.7	69.0	79.6	7.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.6	103.4	75.4	66.5	99.3	74.3	65.3	95.2	73.5	35.5	84.4	34.4	82.2	33.4	80.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.6	11.9	DB	35.3	100.5	2.2	3.1	33.8	102.8	32.5	104.6	31.3	106.4	29.7	108.6	(4)
(5)			WB	34.6	74.6	2.1	0.9	33.1	75.3	31.8	75.8	30.6	76.3	29.1	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	72.6	72.8	70.6	65.8	61.6	55.9	55.3	56.8	59.2	63.6	65.8	70.1	(6)	
	DBStd	7.84	4.43	4.31	4.88	5.60	4.74	4.68	5.57	4.42	4.68	5.33	4.59	4.40	(7)	
	HDD50	18	0	0	0	0	0	4	11	2	1	0	0	0	(8)	
	HDD65	1363	2	2	8	56	127	278	309	258	181	91	44	7	(9)	
	CDD50	5174	700	637	638	474	360	180	174	213	277	422	475	624	(10)	
	CDD65	1043	237	219	181	80	22	4	7	4	7	47	69	166	(11)	
	CDH74	6001	1327	1192	911	498	158	41	87	59	97	320	367	944	(12)	
	CDH80	1772	408	360	264	177	31	12	17	12	30	105	91	265	(13)	
(14) Wind	WSAvg	4.6	6.0	5.4	4.8	4.0	3.4	3.2	3.3	3.5	4.1	5.2	5.8	6.0	(14)	
(15) Precipitation	PrecAvg	17.5	0.9	1.4	1.4	2.2	1.1	1.5	1.5	1.8	1.3	2.0	1.5	1.3	(15)	
	PrecMax	24.1	2.5	2.9	3.9	6.3	2.6	2.4	3.9	7.0	2.7	4.8	5.9	3.9	(16)	
	PrecMin	11.3	0.0	0.2	0.2	0.2	0.1	0.4	0.3	0.2	0.2	0.3	0.2	0.0	(17)	
	PrecStd	3.6	0.6	0.7	0.9	1.7	0.7	0.6	0.9	1.5	0.7	1.5	1.4	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.7	92.6	93.1	92.8	83.6	81.4	82.2	81.0	85.7	90.4	89.2	91.8	(19)	
		MCWB	70.9	70.2	69.5	66.1	60.5	59.3	57.7	58.6	60.2	66.7	66.0	69.2	(20)	
	2%	DB	88.4	87.8	86.2	84.2	78.0	72.6	75.3	73.8	76.4	82.1	82.0	86.3	(21)	
		MCWB	69.6	69.2	68.5	65.6	59.3	55.8	55.4	57.6	59.2	63.9	64.6	68.2	(22)	
	5%	DB	84.6	84.1	82.5	78.6	74.2	68.3	70.4	69.3	72.0	76.7	77.9	82.7	(23)	
		MCWB	68.5	68.3	67.0	63.7	59.1	55.3	54.9	55.8	57.9	61.5	63.3	66.6	(24)	
	10%	DB	81.4	81.0	79.0	74.6	70.7	65.1	65.7	66.0	68.5	72.6	74.6	79.5	(25)	
		MCWB	67.3	67.2	66.0	62.3	58.6	54.5	54.3	55.2	56.9	60.6	61.9	65.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.0	73.0	71.7	68.1	64.3	61.0	59.5	62.1	63.9	68.8	68.6	72.0	(27)	
		MCDB	87.6	88.9	85.6	85.3	74.7	68.2	73.5	72.2	80.8	84.0	82.6	86.8	(28)	
	2%	WB	71.3	71.1	69.9	66.6	62.3	59.1	57.9	59.8	61.2	65.4	66.2	69.6	(29)	
		MCDB	83.6	83.2	83.3	79.3	71.8	68.2	68.9	68.4	72.7	77.4	77.7	82.0	(30)	
	5%	WB	70.1	69.7	68.5	65.1	60.8	57.4	56.6	57.9	59.3	63.4	64.6	68.1	(31)	
		MCDB	80.5	79.5	79.1	75.6	69.9	65.6	65.7	66.2	69.3	73.4	74.9	79.2	(32)	
	10%	WB	68.7	68.4	67.3	63.8	59.5	55.8	55.3	56.2	57.6	61.8	63.1	66.6	(33)	
		MCDB	78.3	77.6	76.2	72.8	68.0	62.6	63.7	63.6	66.4	70.3	72.5	76.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	18.2	18.1	18.7	18.3	18.6	19.5	20.2	18.8	18.1	17.8	18.0	(35)	
		MCDBR	23.4	23.8	23.2	25.4	23.2	24.2	27.5	26.8	25.9	26.0	24.3	24.6	(36)	
	5% WB	MCWBR	8.0	8.4	8.0	9.3	9.3	11.7	11.8	12.1	11.1	10.1	9.5	9.2	(37)	
		MCWBR	20.5	20.7	20.3	22.4	19.6	19.5	22.5	21.3	22.8	21.6	20.9	21.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.360	0.362	0.349	0.326	0.312	0.300	0.303	0.327	0.339	0.361	0.349	0.357	(40)	
		taud	2.489	2.489	2.518	2.570	2.588	2.590	2.573	2.493	2.441	2.413	2.482	2.486	(41)	
	Ebn at Noon		310	303	295	285	272	267	272	279	293	299	311	312	(42)	
		Edh at Noon	36	35	32	27	23	22	23	29	34	38	36	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2414	2159	1760	1310	973	833	916	1177	1600	1974	2316	2484	(44)	
	RadStd		84	74	86	59	51	47	49	50	80	107	95	128	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		N/A	N/A	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page